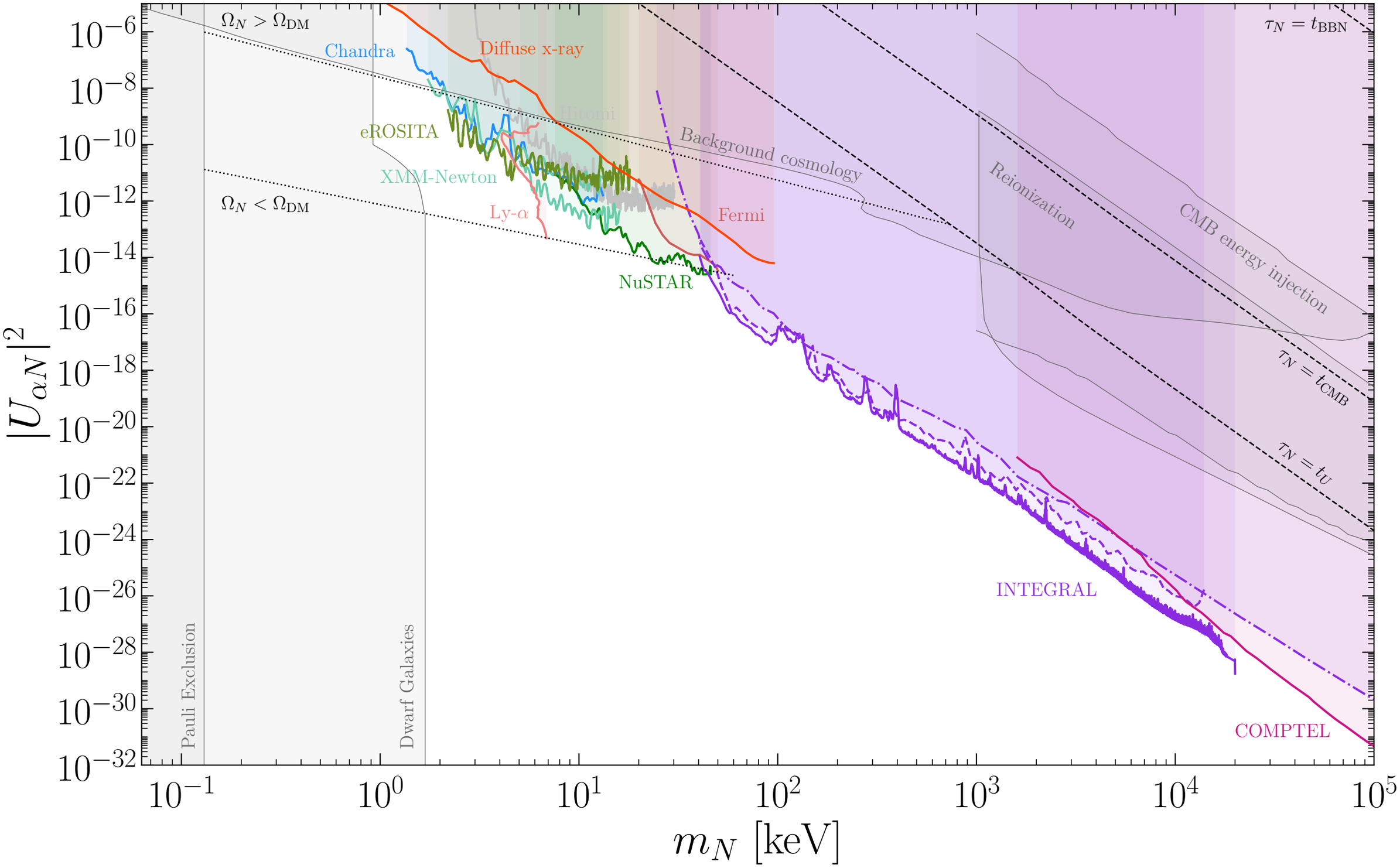

$\Omega_N > \Omega_{\rm DM}$
$\Omega_N < \Omega_{\rm DM}$
$\tau_N = t_{\rm BBN}$
$\tau_N = t_{\rm CMB}$
$\tau_N = t_U$
Chandra
Diffuse x-ray
Hitomi
eROSITA
XMM-Newton
Ly-α
Fermi
NuSTAR
Background cosmology
Reionization
CMB energy injection
INTEGRAL
COMPTEL
Pauli Exclusion
Dwarf Galaxies
$|U_{\alpha N}|^2$
m_N [keV]
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
10^{-18}
10^{-20}
10^{-22}
10^{-24}
10^{-26}
10^{-28}
10^{-30}
10^{-32}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}